THE KIT · WHAT THE MARKETED WORK ACTUALLY REQUIRES

Everything that has to be in the truck, and what it is there for

The equipment needed to deliver the six service lines on the signed agreement — felling, stump grinding, slash and chipping, rounds, defensible space, fuel break and thinning — plus the work that turns up whether or not it was sold. Climbing is the big one of those, and it has its own sheet.

GEAR IS NOT THE GATE

General liability and workers comp are the gate, and nothing on this list substitutes for them. A pull rope does not make felling over a roof survivable if the roof breaks. And **owning ANSI-rated equipment is not a certification.** A helmet that meets Z89.1 does not put the words licensed, insured, bonded, certified or Firewise into a sentence a customer reads. None of them are true yet.

What is already owned

Saws for felling and limbing, a stump grinder, a chipper, haul capacity. Every machine on the sellable service list is in hand. **What is missing is not machines — it is the safety, fire, medical and documentation kit that has to ride with them,** and that is most of what follows. The one machine gap is unchanged and not reopened here: no masticator or mulching head, which is a quoting decision.

The two standards this list is built on

DOCUMENT	WHAT IT IS, AND WHY IT BINDS SIXFOLD
OSHA 29 CFR 1910.266 Logging Operations	**Federal law — enforceable, with penalties.** It defines *logging operations* as felling and moving trees from the stump to the point of delivery, expressly including marking danger trees, felling, limbing, bucking, chipping and transporting. That is a description of a defensible space job. It binds the moment anyone but the owner runs a saw.
ANSI Z133-2026 Arboricultural Operations	**The industry consensus standard — not law in itself,** but what an insurer, an expert witness and a county contractor list measure against. The 2026 edition **took effect 27 May 2026** and supersedes 2017. ISA gives the digital PDF away at no cost — download it.

Where the two differ, this list follows the stricter one and says which.

How the list is organised

- **Sheets 2–5 — buy before job one.** PPE, the cutting kit, the Colorado fire kit, first aid, site control, documentation. Nothing optional, nothing expensive relative to one claim.
- **Sheet 6 — climbing, which is a decision before it is a purchase.** Plus the electrical classification, which is training and cannot be bought at all.
- **Sheet 7 — buy when the job requires it.** Material handling, the trailer, consumables, and what this list leaves out.

No prices anywhere on these sheets. Every figure here traces to a standard or a statute; not one price would, and a plausible invented number is exactly the failure this project has already been bitten by. Price the list at a real supplier and record what it actually cost.

BUY BEFORE JOB ONE · 1 OF 4

PPE, counted per person on site — not per crew

A visitor, an estimator and the owner standing in a drop zone need the same protection as the person holding the saw. Buy sets. The standard numbers are worth writing on the purchase order, because a supplier will happily sell a hard hat that is not Class E and chaps that meet one of the two required standards.

THE LIST, AND THE STANDARD NUMBER TO PUT ON THE ORDER

ITEM	STANDARD	WHY IT IS ON THE LIST
Helmet / hard hat	ANSI Z89.1 Class E	Z133 3.3.4 requires head protection for all arboricultural operations; 1910.266(d)(1)(iv) requires it wherever objects can fall or fly. **Class E** because Z133 requires it near electric supply lines and residential lots have service drops.
Eye protection	ANSI Z87.1	Z133 3.3.6 and 1910.266(d)(1)(vii)(A).
Face protection mesh logger screen	ANSI Z87.1	1910.266(d)(1)(vii)(B) requires face protection where facial injury is possible and **names chipper operation**. Logger mesh screens are permitted for saw work. Where face protection covers both eyes and face, the separate eye device is not additionally required.
Hearing protection	–	Z133 3.3.5 above an 8-hour TWA of 85 dBA. A saw and a chipper both clear that comfortably. Muffs mounted on the helmet is the version people actually wear.
Chain-saw leg protection chaps or trousers	ASTM F1897 *and* F1414	**Z133 3.3.8 requires both standards for ground saw work. 1910.266(d)(1)(vi) separately requires cut-resistant leg protection covering the full length of the thigh to the top of the boot on each leg.**
Chain-saw boots	ASTM F1818	1910.266(d)(1)(v): foot protection that covers and supports the ankle, waterproof or water-repellent, and **cut-resistant for saw operators**.
Work gloves	–	Mandated by 1910.266 specifically for handling wire rope. Everything else is judgement. Buy them anyway.
High-visibility vest	ANSI/ISEA 107	Z133 3.5.3 for anyone exposed to moving roadway traffic. Cul-de-sac work triggers this more often than it looks.

THE ONE EXCEPTION WORTH KNOWING

1910.266(d)(1)(vi) does *not* require leg protection for a climber where the employer can demonstrate that wearing it poses a greater hazard in that particular situation. It is a real exception, not a loophole: the burden of demonstrating it sits with the employer, and it does not extend to anyone standing on the ground.

COUNT IT OUT — ONE COLUMN PER PERSON

- [ ] Helmet, Z89.1 Class E, with chin strap
  Chin strap matters the moment anyone leaves the ground
- [ ] Z87.1 safety glasses
  Second pair in the truck; they get lost, not broken
- [ ] Chaps or saw trousers, F1897 *and* F1414
- [ ] Gloves
- [ ] Spare set of chaps in the truck
  Chaps that have taken a hit are finished, not repairable — and the job stops without them
- [ ] Mesh face screen fitted to the helmet
- [ ] Helmet-mounted ear muffs
- [ ] Saw boots, F1818, ankle support
- [ ] Hi-vis vest, ANSI/ISEA 107
- [ ] Sun and cold layers for a Front Range winter
  Dormant season is the sell; it is also January at 7,000 ft

BUY BEFORE JOB ONE · 2 OF 4

The cutting kit, and the ropes that decide where the tree lands

Saws

- [ ] **Felling saw**
  Bar matched to the largest stem actually being sold, not the largest tree imaginable
- [ ] Two or more spare chains per bar
  A dull chain is the commonest cause of a bad cut
- [ ] Scrench and the saw's own tool kit
- [ ] Bar oil, and fuel in **approved, clearly marked containers**
  Z133 3.6.4. And 3.6.2: fuel with the engine shut down, from approved containers, **at least 10 ft from any operating equipment**
- [ ] **Limbing saw** — a lighter second saw
  Two saws is not luxury. It is what stops one pinched bar ending the day
- [ ] Files, file guide, depth-gauge tool, stump or bench vise
  Chains get touched up at the truck, not at the shop
- [ ] Spare plugs, air filters, bar nuts, starter cord
- [ ] Working **chain brake** on every saw
  1910.266(e)(2)(i) requires one and forbids disabling any kickback device. Z133 6.4.3 prohibits drop-starting outright

TWO SAW RULES THAT ARE EQUIPMENT, NOT HABIT

The **muffler stays on and the spark arrester stays in** — see sheet 4, where in Colorado that stops being maintenance and becomes a condition of running the saw at all. And the chain brake is engaged or the engine shut off **any time the saw is not actively cutting** — before starting, setting it down, or moving both feet on the ground (Z133 6.4.6).

Felling and directional control

- [ ] **Plastic felling wedges**, several, three sizes
  Z133 9.4.10 — wedges stop the bar binding. Plastic only, near a chain
- [ ] Felling lever / cant hook
  For stems too small to wedge and rounds too heavy to roll
- [ ] **Throwline, throw weights, cube bag**
  The item that most often removes the need to climb at all
- [ ] Dead-blow hammer or splitting maul to drive them
- [ ] **Pull rope, long enough to clear the arithmetic below**
- [ ] Come-along or hand winch, slings, shackles, block
  Z133 9.4.9: where a piece falling wrong would damage property, a rope, block, come-along or winch **shall** be used. On a lot with a house, a fence and a neighbour, that is most trees
- [ ] Flagging tape, marking paint, axe or hatchet
  The agreement asks how removals were identified — *flagged, painted, walked and agreed.* This is the gear that answers it

THE ROPE HAS TO BE LONGER THAN THE STANDOFF, OR SOMEONE STANDS WHERE THEY SHOULD NOT

Z133 9.4.5 sets the geometry of a manual felling operation. Workers **not** involved stay back **at least two times the height** of the stem. Workers who **are** involved — ropes, work lines, come-alongs, winches — stay back **at least one-and-a-half times the height**. Both hold until the saw operator says it is safe to approach. Buy rope against that number, not against the tree.

ALSO REQUIRED BEFORE THE CUT, AND FREE

- **A planned retreat path**, cleared — 9.4.6. For the saw operator it runs roughly **45° either side of a line opposite the intended fall**.
- **A drop zone, established before anything is dropped or lowered**, with people excluded from it — 3.4.11.
- **A job briefing before each day or shift** — hazards, procedures, assignments, PPE — 3.4.3. It should be documented, and that document is the cheapest evidence of competence Sixfold can generate.
- **A tree inspection including the root collar**, before anyone works on it — 3.4.5.

BUY BEFORE JOB ONE · 3 OF 4

The fire kit is regulatory in Colorado, and the medical kit is not the one in the catalogue

Fire — the sheet most out-of-state checklists get wrong

Under **Stage 1 fire restrictions**, running a chainsaw is *prohibited* unless all three of these are present: a **USDA Forest Service or SAE-approved spark arrester** properly installed and working, a **chemical fire extinguisher**, and **one round-point shovel**. Not recommended — a condition of being allowed to cut.

BUY PAST THE DISAGREEMENT

The order read first-party for this sheet — BLM Fire Prevention Order **COG02-25-01** — specifies an extinguisher of **not less than 8 oz by weight** and a shovel of **at least 36 inches overall**. Secondary summaries of Colorado Stage 1 restrictions say 35 inches. Rather than settle it, buy past both: a **5 lb 2A:10B:C extinguisher** and a **36-inch-plus round-point shovel**. Orders are issued jurisdiction by jurisdiction and their wording varies — read the one in force for the county you are working.

- ☐ **Extinguisher on every machine and every vehicle**
  1910.266(d)(4), and it applies with no fire restriction in force at all. Not one shared extinguisher in the truck
- ☐ **Spark arresters on everything with an engine**, checked
  Z133 3.6.10: SAE J335 or USDA FS 5100-1. They clog
- ☐ Fuel spill kit — absorbent, rag bag
  Z133 3.6.3 requires spills cleaned up or protected from ignition; 3.6.7 requires fuel-contaminated clothing changed
- ☐ **Round-point shovel, 36 in or longer**
  Lives with the saw. Does not get borrowed for the stump job
- ☐ Backpack water pump, or at minimum a 5-gallon can
  Not named in the orders read. It is what actually kills a chip-pile smoulder
- ☐ **Check the restriction stage before every job in fire season**
  Stage 2 can prohibit chainsaw use outright in some windows. A scheduling fact as much as a gear one — put it on the job briefing

First aid and emergency

- ☐ **First-aid kit meeting ANSI Z308.1** — at least two
  Z133 3.2.2 sets the standard. 1910.266(d)(2) sets the placement: one at each site where trees are being cut, one at each active landing, and one on each transport vehicle
- ☐ Charged phone, and a plan for no signal
  Z133 3.2.1 requires everyone instructed in 911 and other emergency calls. Foothill lots lose signal
- ☐ Eye wash, burn gel, tick remover, insect sting kit
  Z133 3.2.3 expects instruction in poisonous plants, stinging and biting insects
- ☐ **A bleed-control kit on top of that**
  Tourniquet, pressure dressing, packing gauze. A saw leg wound is an arterial bleed and the Z308.1 minimum is not built for it. The cheapest life-saving item on the page
- ☐ The property address written down
  Including gate code or mile marker where there is no house number. The dispatcher will ask
- ☐ Water, more than anyone thinks
  Front Range summer, altitude, PPE

Check the kit against Appendix A to 1910.266, which is mandatory, rather than against whatever a supplier calls a logging kit: gauze pads including two at 8×10 in, a roller bandage, two triangular bandages, a wound-cleaning agent, scissors, tweezers, adhesive tape, adhesive bandages, latex gloves, resuscitation equipment, two elastic wraps, a splint, a blanket, and written directions for requesting emergency assistance.

ONE ITEM ON THIS SHEET CANNOT BE BOUGHT

Two people trained in first aid and CPR shall be available wherever two or more workers are at a work location — Z133 3.2.5. There is an exception allowing one where every new hire is trained **within three months of their hire date**, which is the practical route for a company this size. Either way it is a hiring-plan item, and it wants scheduling now rather than the week before spring.

BUY BEFORE JOB ONE · 4 OF 4

The site kit, the locate, and the box of paper that is the whole positioning

Site control and the neighbours

- [ ] Cones and caution tape
  For the drop zone Z133 3.4.11 requires established before anything is dropped or lowered
- [ ] Plywood or track mats for a driveway
  If the chipper or grinder crosses it
- [ ] Rakes, chip shovel, pitchfork, stiff broom, leaf blower
- [ ] Tarps
  Lawn protection, chip control, gathering small debris. The cheapest thing that makes a finished job look professional
- [ ] **White marking paint**
  For the 811 pre-mark, below
- [ ] Contractor bags, baling twine, zip ties, duct tape

CALL 811 BEFORE ANY STUMP GRINDING — IT IS EXCAVATION

Colorado 811 lists **tree root removal and stump grinding** as excavation requiring a locate. The statute, **C.R.S. 9-1.5-103(3)(b)**, sets notice at **at least two business days, not counting the day of notice** — but Colorado 811's own excavator page says two in one place and three in another. **Plan on three business days, and pre-mark the area in white paint, flags or stakes** as they require. Z133 5.6.4.4 puts the same duty on the grinder operator, and the practical consequence is scheduling: a stump job cannot be sold as same-day.

The documentation kit

This is the part that makes Sixfold Sixfold, and it costs less than a chain. Every funding channel in the positioning wants evidence, and the evidence has to be captured on the day — nobody goes back for a before photo.

- [ ] **Phone or camera, for same-angle before and after**
  Fixed positions, shot from the property line and from the house. The portfolio, the insurer's documented-mitigation evidence, and the tax credit photograph log, all from one habit
- [ ] **The four tax credit client forms**
  Eligibility record, work and cost breakdown, claim file checklist, photograph log — from *tax-credit-notebook/*
- [ ] **Measuring wheel or long tape**
  Defensible space is sold in distances — the first five feet, firewood at 30 ft, zone boundaries. A written scope that names a distance needs it measured
- [ ] Business cards and the QR code card
  linktr.ee/SixFoldColorado — print black version, verified to decode down to 160px
- [ ] **The tablet or printed agreement pack**
  Service agreement, both cancellation notices, and the internal field crib sheet — from *agreements/*. The crib sheet never gets handed over
- [ ] **The funding one-pager, in quantity**
  It is the campaign. It survives the door closing, which no pitch does
- [ ] Weatherproof clipboard, pens that work cold, a folder that closes
- [ ] The property log
  Address, outcome, what was seen in the yard. That log is next spring's target list

ONE DOCUMENT, THREE USES

The written scope on every job — what was cut, where, to what spacing, what happened to the material — becomes the county's scope-of-work format, the insurer's documentation, and the homeowner's tax-credit receipt trail. It is the item on this sheet most likely to get skipped when the crew is tired, and it is the entire point of phase 1.

WORKING ALOFT · NOT A SHOPPING LIST

Climbing is a decision before it is a purchase

Nothing Sixfold markets requires climbing — defensible space, thinning and fuel breaks are ground work. It arrives when a hazard tree cannot be dropped whole, over a roof or a fence, and that is a real fraction of residential jobs.

OPTION	WHAT IT COSTS
Buy the gear and climb	Cheapest per job, most expensive every other way. Needs training, and a second trained person on site every time.
Rent or hire an aerial lift	Removes the climbing, adds Z133 §5.6.1 entire — and a lift cannot reach many foothill back yards.
Subcontract the climbing removals	Keep the ground work, the chipping, the grinding and the paperwork; give away the margin on the climb. **The phase 1 answer, until GL and comp are in force and someone is trained.**

THE RULE WITH TEETH, BEFORE ANY OF THE GEAR BELOW

Z133 3.4.7 and 8.2.2: a second arborist, trainee, or worker **trained in emergency procedures** shall be within visual or voice communication during any arboricultural operation **above 12 ft (3.65 m)**. **One person cannot climb.** That is a staffing cost, not an equipment cost, and most of the reason the recommended option is the third. Ladders (§7.5) do not escape it. And 3.2.4 requires anyone who may face a rescue decision to be trained for it — Annex F is the flowchart.

If and when Sixfold climbs — Z133-2026 §8 is prescriptive

ITEM	SPECIFICATION
Climbing line and work-positioning lanyard	Minimum **0.44 in (11 mm)** diameter · synthetic fibre · **5,000 lb (22.24 kN)** breaking strength, new and unterminated · elongation not over **7% at 500 lb** · marked by the maker as suitable for tree climbing — 8.1.5
Saddle	Marked to a governing standard — **ASTM F887** if manufactured after 1 January 2018 — 8.1.3
Carabiners	Self-closing **and self-double-locking**, gate needing **two consecutive deliberate actions** to unlock, **5,000 lb** major axis — 8.1.9
Snap hooks · screw links · Prusiks and hitch cords	All **5,000 lb** minimum, new and unterminated — 8.1.6, 8.1.8, 8.1.11
A second means of being secured	A climbing line **and at least one other means of being secured**, carried at all times — 8.2.4 — and **both in use** while running a saw aloft — 6.4.5
Hand saw	**Required, not optional** — scabbarded or folding — 8.2.5
Climbing spurs	Gaffs suited to the tree, sharpened to the manufacturer's schedule — 8.1.16. **Removals only** — spurring a tree that stays is damage
Throwline and weights · top-handle saw · rigging rope, blocks, friction device	Set the line from the ground, inspect the anchor before loading it — 8.2.11. Top-handle saws are for a trained climber aloft, nowhere else. And 9.4.7 first asks whether the tree can take the lowering loads at all

Electrical — the item that is training, and cannot be bought

There is a service drop over most Front Range driveways. Z133 §4 sorts arborists into four electrical levels, and **an untrained crew is Electrical Level 1 — unqualified for any voltage.** A Level 1 arborist shall not work where electrical hazards exist, shall not approach within the Table 1 minimum approach distance, and **shall not cause vegetation to violate it either.** No purchase changes that. The gear implication is nil; the operational one is large: **look up during the job briefing, and decline the tree that is in the wires.** Declining it costs one job; taking it costs the company.

BUY WHEN THE JOB REQUIRES IT

Material handling, the trailer, and the shelf that decides whether a crew works a full day

Around the machines already owned

- [ ] Skidding or log tongs, timber jack
  Moving rounds without a machine
- [ ] Ratchet straps, cargo net, chains rated for the load
  To rated tie-down points, not convenient ones
- [ ] Trailer lights, spare, jack, chocks, breakaway cable checked
- [ ] Wheelbarrow, tarp system or chip box for the trailer
- [ ] Splitting maul — a hydraulic splitter only if firewood becomes a line
  Softwood cordwood in the Denver metro is margin recovery on wood already on the truck. Do not buy a splitter to chase it
- [ ] **Seat belts used**
  1910.266(d)(3) requires them securely and tightly fastened for machine and vehicle operators

CHIPPER AND GRINDER — THREE THINGS TO MEASURE ON THE MACHINES SIXFOLD ALREADY OWNS

Z133 5.6.2 sets an infeed geometry: a hopper **not less than 85 in** from the blades — or the feed-roller pinch point — to ground over the hopper centre line; a **flexible anti-kickback device** on hand-fed rotary chippers; and on mechanically fed machines a **quick-stop and reversing bar reachable across the top and both sides**. Nobody reaches past the plane of the hopper while anything turns, and panels stay closed until everything has stopped and **the key is pocketed**.

For the grinder — 5.6.4: guards in place whenever the wheel turns, everyone clear, the operator at the controls until it stops, and eye and hearing protection for *everyone* at the site.

The maintenance shelf

- [ ] Bar oil, two-stroke oil, fuel, labelled approved containers
- [ ] **Chipper knives and a spare set, plus the anvil clearance gauge**
  Dull knives are how chippers come to be fed dangerously
- [ ] Grease gun, hydraulic fluid, funnels
- [ ] Spare chains, files, rim sprockets, bar nuts, plugs, filters
- [ ] **Stump grinder teeth, pockets, and the wrench**
  Consumables, and they go fast in Front Range decomposed granite
- [ ] Replacement PPE — a running cost, not a one-off

FLAGGED, NOT ANSWERED — ASK THE BROKER AT THE SAME TIME AS THE GL QUOTE

A truck plus a loaded chipper often clears **10,001 lb combined gross vehicle weight rating**, where federal motor carrier rules begin, with a separate Colorado intrastate regime alongside. Whether Sixfold's combination crosses it — and so whether a USDOT number, markings, driver qualification files or medical certificates apply — **was not researched for this checklist and should not be guessed.** Weigh it. Free to ask, expensive to discover at a roadside inspection.

What these sheets deliberately leave out

- **Mastication.** No masticator or mulching head, by decision. Rent-versus-subcontract is a phase 2 call.
- **Prescribed and pile burning** — a different permit regime, county by county, and not a service Sixfold sells.
- **Cranes, knucklebooms, aerial devices, pesticide application** — Z133 5.6.6, 5.6.1 and §10. Whole classes of equipment and licensing, none of it in the plan. And **prices**, per sheet 1.

Sources. OSHA 29 CFR 1910.266 and Appendix A · ANSI Z133-2026, ISA digital edition · BLM Fire Prevention Order COG02-25-01 · C.R.S. 9-1.5-103 and colorado811.org. Still open: the GVWR question above, municipal tree-service licences, the chipper's infeed geometry, and prices. Fact-check log in gear-checklist.md.